

AI in math is going exponential

A working mathematician's view

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Views my own, not of my employer.

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Who I am, why this talk

- Mathematical physicist: Application of differential geometry to integrable systems, separation of variables.
- Not a frontier ML researcher, just user of public models.
- Today: what a working research mathematician makes of the Erdős news, and why it might matter for AI safety.

Four years of using AI for math

ONE MATHEMATICIAN'S VIEW FROM THE INSIDE

- 2022–23. Couldn't reliably convert Maple $\leftrightarrow$ LaTeX. I gave up on it.
- Summer 2024. Came back, mostly for smoothing prose.
- 2025. Steadily more useful. IMO gold in spring, stronger models in autumn.
- December 2025. Claude Code arrives. Second ChatGPT moment. I become a heavy user.
- May 2026. The Erdős problem news.

The steps get bigger and arrive faster. The Erdős case is the next point on the exponential curve, not a break from it.

The news

The announcement that started this talk

20 MAY 2026

OpenAI says: an unreleased internal model produced a new construction for a long-open problem in geometry.

The prompt and the paper are public on OpenAI's site. Tim Gowers (Fields medalist) and eight other mathematicians verified it, and wrote the final article.

The standard public takeaway: “AI proved a famous problem.” The interesting question is *what kind of work* the model did.

Sources: OpenAI announcement and proof, openai.com (20 May 2026). Reviewers' article: Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimerman, Wang, Wood, [arXiv:2605.20695](https://arxiv.org/abs/2605.20695).

The problem Erdős posed in 1946

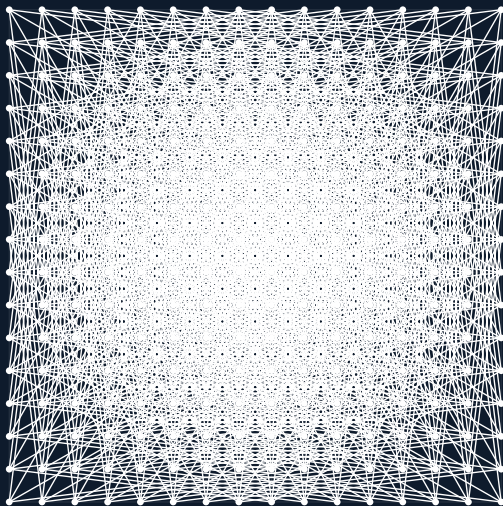
EASY TO STATE, HARD TO SOLVE. OPEN FOR 80 YEARS.

Given n points in the plane, how many pairs can be at exactly unit distance?

More technically, how much does it grow with n ?

Erdős's conjecture, standing for decades: the $\sqrt{n} \times \sqrt{n}$ square grid is essentially optimal.

Easy to state, combinatorial, geometric.



The model beats the grid

THE REVIEWERS' VERDICT

A construction that **outperforms the square grid**. Reviewed by Tim Gowers (Fields medalist) and eight other mathematicians.

“If submitted by a human, I would have recommended publication in *Annals of Mathematics* without hesitation.”

Tim Gowers

What actually happened

IN THE PUBLIC RECORD

Setup. OpenAI tested models against Erdős's open-problem list. The problem was **human-curated**, not the model's own pick.

AI generated prompt. Set the problem up and "Prove or disprove."

What the model did. Wrote the formal problem statement. Produced the proof in a single run. Ran it through an AI grader, which flagged **high confidence**, before any human looked.

Then the humans came in. Verified, simplified, sharpened. The preprint they posted (arXiv only, not journal-published) is human-edited prose.

What we don't get to see: the proof came in one run, but not how many Erdős problems they tried before this one paid off.

What the nine reviewers keep coming back to

THREE THINGS, AGAIN AND AGAIN

One. The model chose to **disprove** the conjecture. The prompt only said “prove or disprove”. The standing belief said prove.

Two. It pulled in tools from **a completely different branch of math**, number theory, that no one working on this geometry problem had thought to use.

Three. The model writeup was at **publication level**. Gowers said *Annals* without hesitation.

The first two are about **picking the direction**. That’s the part we usually call “the mathematician’s job.”

Not just one model, not just one lab

GENERAL-PURPOSE MODELS, POINTED AT THE PROBLEM

Tsoukalas, Chaudhuri et al. (arXiv:2605.22763, 21 May 2026.) An autonomous AI agent works through the list and resolves **9 of 353 open Erdős problems**, plus 44 of 492 from a second famous problem list.

General-purpose models, too. Mythos and ChatGPT 5.5 reportedly also found the construction when pointed at the problem, not just the internal models.

The Erdős preprint is not one lab's stunt. It's something the **current frontier** can do, given the right problem.

Perspective from my field

Where they help, day-to-day

MY DESK, THE LAST TWO YEARS

Sifting literature. What's new in a paper, how it applies to my work. What to read on a topic, especially outside of your expertise.

Writing. Maple output to LaTeX. Paper draft to slide deck. Table of results to prose. Also writing Maple itself: the Haantjes-torsion routine on my software page was generated by an LLM, optimising the naive version I had.

Review. Catching errors in code, calculations, or prose. A different model than the one that wrote it is often the most useful reviewer.

Real hours saved. I am not giving them up.

But the wins are jagged

CAPABILITIES ARE UNEVEN

Maple code gets hallucinated more often than mainstream languages. Probably less represented in the training data.

A full proof of a minor proposition, and then a trivial error in the next message. One I had pointed out three times in the project notes.

Course-correction over more than a few turns reliably produces a long, incoherent wall of text.

Last week: half a day trying to get Claude to adapt a copied Maple routine to a new setting (larger manifold, different coordinates). I gave up.

Where they really fail (so far)

THE THING THE ERDŐS CASE IS ABOUT

LLMs have not yet helped me get unstuck.

Once I have picked the direction, they help with most of what follows. But choosing the direction, the stalled moment when I do not know which branch to follow, is the one thing they have never done for me.

The Erdős case is different. There a model did exactly that: it chose to **disprove** the conjecture, against the standing belief, and reached into number theory to do it. That is the direction, not the work that follows it.

The pattern I had identified

FROM FOUR YEARS OF LLMS IN MY OWN WORK

Precedent in the literature, even obscure: the model finds it.

Right move not in any literature: hallucinations dominate.

AI does the technical work after I pick the direction. It does not pick the direction.

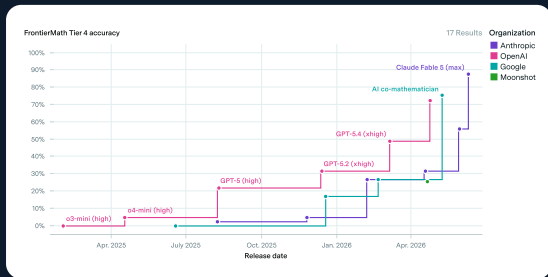
The Erdős case is the first prominent, publicly verifiable instance of it doing the second.

The exponential

The benchmark nobody expected to fall

FRONTIERMATH TIER 4: 50 PROBLEMS, EACH A SHORT RESEARCH PROJECT

Written by professors and postdocs to outrun the models; some were expected to resist AI for decades.



Source: Epoch AI, FrontierMath Tier 4 accuracy (CC-BY), epoch.ai.

Near 0% in early 2025, ~87% a year later, across labs. The same exponential, now at the hardest end.

The autonomy curve

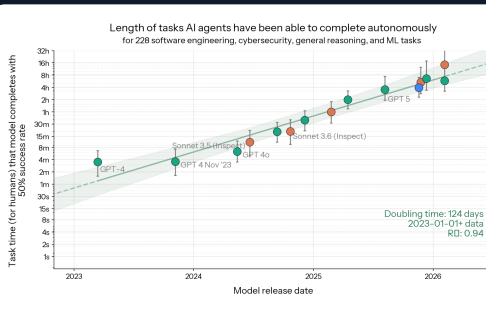
METR: AI ABILITY TO COMPLETE LONG TASKS

Task length an AI finishes alone keeps **doubling**: ~ 7 months over six years, **faster** lately.

Erdős is the **research-grade end** of this curve.

It ran on an **unreleased frontier**, ahead of the model I rent.

Not a sudden discontinuity, a milestone on a curve we already had data on. Expect more like it, and faster.



Source: METR, *Measuring AI Ability to Complete Long Tasks* (CC-BY),
metr.org.

The two-day paper era

THE OTHER EXPONENTIAL

Math papers posted to arXiv, year on year: +32%, a record (4,507 $\rightarrow$ 5,924 in May).

Paata Ivanisvili (UCI), on the new workflow: “Day 1: solve the problem. Day 2: verify, write, submit.”

Capability, autonomy, and now raw output: the same curve wherever you measure. A blunt count, but it moves with the rest.

What the field told itself, one week ago

TWO JUNE 2026 DOCUMENTS FROM THE MATHEMATICAL MAINSTREAM

Leiden Declaration (IMU-endorsed, 1,500+ signatories incl. Tao, Scholze). To policymakers: “don't believe the hype.” Consent, attribution, plausible-but-wrong proofs, public compute independent of industry. And: agenda-setting should stay human.

Nature Opinion piece (He, Graepel/DeepMind, 11 June). Human-AI “partnership”. Draws the line higher: “the decisive creative leaps are still made by humans”, meaning a new *paradigm* (Einstein, Feynman). Grants the Erdős disproof “perhaps the first major result produced by a machine”, yet still files it below those leaps.

Both still place the human part in picking the direction, and both must defend it: one asserts agency should stay human, the other lifts the bar to a new paradigm. You shore up a line, and raise it, only when it is starting to give.

Closing

BEYOND MY OWN DESK

The Erdős case is the first publicly verifiable instance of AI doing what a research mathematician does: picking the direction, bringing in tools from another field, writing it up to the standard of a top journal.

I think that, if the exponential holds, the value of what we now call a mathematician's work may not survive the decade. I'd rather be wrong about that.

And these were general-purpose models, not a math-specific engine. Math and cybersecurity are just the fronts where the work is easy to check; the same progress lands in less visible, more dangerous domains like bioengineering and autonomous weapons.

Why this is the dangerous

TWO WAYS THE SAME CAPABILITY CUTS

Misuse. A **general** model that picks the research direction in math is the same system pointed at bioweapons, cyber-offense, autonomous weapons.

Loss of control. The METR curve measures **autonomy**; “picking the direction” is **initiative**. Scale both, and you get systems pursuing goals we did not set, in domains we cannot check fast enough to step in.

The capability that impressed Gowers is the same capability that makes these systems **hard to oversee and hard to control**.

We can still stop it

THE EXPONENTIAL IS A CHOICE, NOT A LAW

The curve we have been looking at is the result of decisions made in a handful of labs, in a handful of jurisdictions. Every one of those decisions is reversible, and none of them is a law of nature.

Nuclear weapons, human cloning, and gain-of-function research have all been paused, regulated, or banned when the stakes became clear enough. **AGI can be too.**

PauseAI is asking for pausing the frontier model development, until we know we can do it safely. Help us by signing the petition, finding your local group (PauseAI en Español), talking to colleagues about what is actually happening.

Thank you.

Q&A



Join PauseAI: pauseai.info/join